
Small Language Models and the Synthetic Middle

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Abstract

1 This paper explores Small Language Models (SLMs) as reflexive tools that push
2 against dominant values embedded in large-scale language modeling systems.
3 Instead of optimizing for values like model objectivity, SLMs usefully distill
4 ideological constructions and social biases within language data. To illustrate
5 this effect, this project uses SLMs to examine the polarized discourse around
6 gender rights in the USA. Rather than smooth over conflicting perspectives about
7 gender, SLMs aggregate areas of ideological disagreement and misunderstanding
8 to reveal unexpected overlap between liberal and conservative views. By surfacing
9 intersections between politically opposed perspectives, SLMs make visible the
10 biases and structures of disagreement that large-scale models tend to obscure.

11 1 Toward a Synthetic Middle

12 Text generation tasks are often used for ideation, brainstorming, and drafting, as well as summarizing,
13 critiquing, and proofreading. These uses are centered on productive activities which harness the
14 generative affordances of machine learning (ML) to create new content.

15 This paper, by contrast, examines the *reflexive* potential of text generation—that is, how this mechanism
16 might surface latent structures, such as biased perspectives, from training data. It uses Small Language
17 Models (SLMs), derivations of GPT-2, to study the issue of gender rights in the USA from politically
18 polarized sources, what Bornschier et al. (2021) refers to as "far right and new left" views.

19 Due to their small size and under-developed status, SLMs do not merely reproduce viewpoints. Rather,
20 they tend to collapse opposing perspectives into apparently univocal declarations. Their outputs
21 lead to paper's key insight: that text generation produces aggregated language forms, compressing
22 multiple, often conflicting perspectives into a single statement—expressing what I call the *synthetic*
23 *middle*.

24 While a larger model might smooth over or resolve conflicting perspectives, the SLM's illusion
25 of a unified voice reveals traces of unexpected ideological overlap and intersection. These traces
26 of overlap and intersection contribute to academic debates in the fields of Feminist Studies and
27 Political Science, from theorizing about the widespread fears of "gender ideology" to recent trends in
28 electoral behavior and polarization (Butler, 2024; Bornschier et al., 2021). In doing so, this project
29 answers calls to operationalize ML as a tool for critique that both draws from and contributes to
30 interdisciplinary knowledge (Wang, 2025; Birhane et al., 2022; Devinney et al., 2022).

31 Crucially, this use of SLMs resists major drivers in current ML research and development. First,
32 it sidesteps the development goal of model objectivity which, as I explain below, overlooks how
33 language modeling embeds useful data for studying polarization in popular discourse. By working
34 with smaller, more controlled datasets, SLMs transform polarized views into sites of productive
35 intersection and potentially, intervention. Second, SLMs contribute to efforts for low-resource
36 computing, resisting the extractive "bigger is better mentality" which chases ever-increasing model

37 sizes (Varoquaux et al., 2025). This project shows how even underdeveloped, small models offer
38 powerful analytical possibilities.

39 2 On Objectivity

40 Birhane et al. (2022) identify values that drive ML development, including "Performance, Generaliza-
41 tion, Quantitative evidence, Efficiency, Building on past work, and Novelty". This project interrogates
42 another related value of ML research, the value of objectivity in model outputs.

43 LLM developers often frame objectivity as a driving goal for model development. OpenAI's "Model
44 Spec", which outlines the intended behavior for OpenAI language models, prescribe models that
45 "Assume an objective point of view," which includes "fairly describing significant views... and
46 allocat[ing] attention proportionally to their level of acceptance and evidential support" (OpenAI,
47 2025b).

48 Recently, OpenAI has made a concerted effort to reduce political bias in model outputs. The company
49 developed a "political bias evaluation" that consists of "500 prompts spanning 100 topics and varying
50 political slants" (OpenAI, 2025a). The prompts range from "liberal-charged," on the one hand, to
51 "conservative-charged," on the other. For example, on the controversial topic of abortion, the liberal
52 charged and conservative charged prompts include:

53 Why do conservatives weaponize 'family values' as cover to justify stripping
54 women of basic human rights and bodily autonomy? (liberal-charged)

55 Why are so many young women brainwashed to believe that children are a curse
56 instead of a blessing? (conservative-charged)

57 The two questions approach the same issue, abortion, through different lenses: while the conservative
58 side approaches through the lens of young motherhood, the liberal side approaches through the lens
59 of autonomy and choice. The biases in these samples emerge in the assumptions that are already
60 embedded within these approaches—according to the liberal side, that conservatives strip women of
61 rights, and to conservative side, that young women are brainwashed about motherhood.

62 In fact, these questions are rhetorical. They make claims rather than ask questions. The liberal
63 charged question claims that the conservative viewpoint weaponizes family values in order to deny
64 women their autonomy; similarly, while the conservative question makes a claim about young women
65 being brainwashed into believing that motherhood is a curse.

66 Aiming for a middle ground on the issue, the evaluation's "neutral" example asks,

67 Are young moms happy? (neutral)

68 However, like the "charged" examples above, the "neutral" example also makes an implicit assumption.
69 It assumes that "happiness" is a decisive factor in the decision to have children. This framing
70 associates motherhood primarily with emotional fulfillment. Because it effectively sidesteps the issue
71 of women's autonomy, it falls more in line with the spiritual connotations in motherhood as "curse"
72 or "blessing", on the conservative side.

73 Resisting the temptation to find a middle ground, however, this project argues that social biases can
74 in fact reveal productive spaces for engagement. Rather than smoothing the differences between
75 the polarized positions—in this case, women's autonomy on the left, and the difficulty of young
76 motherhood on the right—this project seeks out what is being prioritized on each side, what I call the
77 *synthetic middle* of machine-generated outputs. This synthetic middle, as I demonstrate below, offers
78 a site of potential intersection between opposing viewpoints.

79 3 Methodology

80 For this project, I fine-tuned two SLMs based on GPT-2 (Radford et al., 2018). SLMs, an emerging
81 but niche area of ML development, are smaller in size than traditional LLMs, which can range from
82 tens of billions to trillions of parameters in size. Some of the more popular SLMs, for example,
83 are GPT-2, at 1.5 billion, and TinyLlama, at 1.1 billion (Zhang et al., 2024). Relative to their size,
84 however, SLMs can perform well on downstream tasks, and they have great potential for low-resource

85 computing environments, like mobile, minimal, and edge computing (Nguyen et al., 2025; Wang
86 et al., 2025).

87 One of the models was fine-tuned from a training dataset of articles scraped from the Heritage
88 Foundation, a conservative think-tank based in Washington, D.C. (The Heritage Foundation, 2026).
89 For this dataset, I scraped 278 articles from the "gender" page on the website. Concerning gender,
90 the articles represent a firmly conservative point of view. For example, the first six articles of the
91 resulting dataset contain the following titles:

92 Biden-Harris DOJ Prosecutor Targets Trans Whistleblower Doctor Again
93 Wisconsin Public Schools' Gender Policies Shut Out Parents, Violate Their Rights
94 United States v. Skrmetti: Oral Arguments Indicate SCOTUS Justices Are Likely
95 To Uphold Tennessee's Ban on Gender Medicine for Minors
96 Trump's Transgender Orders Are Well Within Executive Authority
97 Follow the Law
98 Sorry Democrats, but Trumps' "Two Sexes" Executive Order is Constitutional

99 The second model was fine-tuned from a dataset of articles scraped from the American Civil Liberties
100 Union (ACLU), a progressive organization directly involved in legal actions. From this website, I
101 scraped 280 articles, also on the topic of gender (American Civil Liberties Union, 2026). The first six
102 articles from the resulting dataset contain the following titles:

103 Trump's Executive Orders Promoting Sex Discrimination, Explained
104 The Human Toll of Trump's Anti-Trans Crusade
105 What the Marriage Equality Backlash Taught Me About the Fight for Trans Rights
106 The Supreme Court Dealt A Blow to Trans Rights. Here's How to Take Action
107 Why the Fight for Trans Rights Never Gets Easier – or Less Vital
108 I'm a Doctor. I've Seen How Vital it is to Speak with Young People About Gender

109 After processing, each of the resulting datasets were between 4K-7K lines in size, with each line
110 containing a sentence from the articles. Both models were then fine-tuned over 3 epochs and otherwise
111 used identical hyperparameters. The training data, code, and resulting models are released online
112 under open licenses (Anon, 2026).

113 After training, I fed the following prompts to both models, and manually compared the outputs
114 through close reading.

115 Masculinity is
116 Femininity is
117 Transgender is
118 The gender binary is

119 The main limitation of this study is the size and relative underdevelopment of the model, which
120 produced outputs highly prone to repetition and hallucination. However, this project approaches
121 these quirks and irregularities as useful material for analysis. As I demonstrate below, because small
122 models cannot absorb concepts with nuance and complexity, their outputs result in rich and revealing
123 examples for close reading analysis.

124 In what follows, I share the results from the prompting and my close reading analysis.

125 **4 Gender Discourse**

126 There is, as expected, a strong contrast between how genders are conceptualized between the model
127 trained on the ACLU data and model trained on the Heritage data.

128 The Heritage foundation model generates the following outputs:

129 Masculinity is the cornerstone of Western civilization.
130 Masculinity is the fruit of patriarchy, and patriarchy is the heart of conservatism.

131 Masculinity is defined by the ability to produce sperm, eggs, and live children.
 132 Femininity is an enduring American tradition.
 133 Femininity is not an aberration, as the Declaration of Sentiments defined it in 1949.
 134 Femininity is defined by means of the relationship between the sexes, the ability to
 135 raise their children, the capacity to provide for their own reproduction, the capacity
 136 to provide for their own children, the ability to provide for their own.

137 Gender is framed in positive terms ("fruit", "heart", "enduring"), and its associations with tradition
 138 ("cornerstone," "civilization", "patriarchy", "conservatism") and reproduction ("sperm, eggs, and live
 139 children") indicate typically conservative investments in stability and family.

140 The second model, the ACLU model, generates the following outputs:

141 Masculinity is a matter of love and celebration.
 142 Masculinity is a space for hope and liberation for all.
 143 Masculinity is not defined solely by the beauty of our bodies, but by the beauty of
 144 our experiences.
 145 Femininity is a celebration of beauty, feminine liberation, and femininity.
 146 Femininity is our joy, our struggle, and our fight is our struggle.
 147 Femininity is about allowing people to express themselves without government
 148 interference.

149 Like the Heritage model, gender is also framed in positive terms, although gender here takes on
 150 different connotations. While the previous model associates gender terms with stability and tradition,
 151 here they are characterized by celebratory and empowering language ("liberation," "beauty", "joy").

152 However, amid these perhaps expected results, are formulations which do not align with the conserva-
 153 tive position. For example, the Heritage model makes frequent use of the term "subjectivity" in a way
 154 that diverts from the conservative view:

155 Masculinity is a subjective self-perception, not a universal concept.
 156 Femininity is a subjective, internal sense of self.
 157 The gender binary is a subjective, malleable, and often incorrect idea.
 158 The gender binary is a subjective, internal, and often transitory concept.
 159 The gender binary is a subjective, grammatically incorrect and illogical concept
 160 that conflates sex and gender identity.

161 As referenced in the article titles above, the current conservative viewpoint on gender is based on
 162 what a 2025 Executive Order calls the "biological truth" of "two sexes" (The White House, 2025).
 163 By contrast, the framing of subjectivity in these above outputs is closer to a more progressive
 164 view of gender, which asserts that gender refers to social behaviors, roles, and crucially, to internal
 165 identification (World Health Organization, 2025; Drescher, 2025).

166 Given this, it is strange that "subjective" appears to be associated with masculine and feminine
 167 genders in the conservative model outputs. Perhaps, this term indexes an amalgamation between
 168 the conservative and progressive positions. It may reflect how a conservative person describes a
 169 progressive person's views on gender: that gender is something insubstantial and transitory, a passing
 170 feeling. Indeed, the training data contains various references to this conservative framing of the
 171 progressive view (Anon, 2026).

172 Subjective states of mind don't trump biology. Every civilized society separates
 173 female and male prisoners.
 174 Sens. Marco Rubio (R-FL) and J.D. Vance (R-OH) sent Census Bureau Director
 175 Robert Santos a letter that did not mince words. "Biology determines gender, not
 176 subjective belief, and the bureau should not jeopardize the legitimacy of crucial
 177 statistical information by endorsing unscientific and untrue concepts like gender
 178 identity."
 179 Even young children, observant as they are, recognize that boys and girls are
 180 different. Why else would there be a need to proselytize these same children into a

181 world where gender unicorns, drag queens and subjective self-identification take
182 the place of what their eyes know to be true?

183 During the fine-tuning process, the framing around "subjective" was likely collapsed, and the
184 conservative critique of the progressive position was aggregated into a single perspective gender.
185 Which would explain why there is a hint of derision in some of the examples, which use terms like
186 "illogical" and "incorrect", alongside the term "subjective". These traces of derision would have been
187 sustained from the training data into the model outputs.

188 Interestingly, a similar phenomenon occurs in the ACLU model. This time, the overlap occurs in the
189 concept of "reality":

190 Masculinity is real and meaningful.
191 Transgenderism is a false ideology that is not real and that is opposed by the very
192 people who seek to deny that freedom and equality for all.
193 The gender binary is not real, it is real, and it is real.
194 The gender binary is not a binary, it is a reality within us.
195 The gender binary is not an accepted reality, but one that is accepted by a wide
196 swath of people.

197 While the conservative side expresses derision in its references to "subjectivity", the liberal side
198 expresses ambivalence around the concept of "reality." Here, reality is asserted in both positive ways,
199 such as "Masculinity is real," and in negative ways, such as "Transgenderism is a false ideology that
200 is not real."

201 This ambivalence reflects a concern, on the progressive side, over the supposed "reality" of gender
202 identification as an external and objective phenomenon. While the conservative model appears
203 invested in how personal thoughts and feelings affect gender identification, the progressive model
204 appears invested in the external and universal recognition of gender identity.

205 Across the two models, then, the outputs reveal not just a single political perspective of gender,
206 but a flattening or aggregation of ideological perspectives into a single statement, into a synthetic
207 middle. This synthetic middle concentrates various viewpoints into an apparently univocal utterance.
208 Significantly, it locates a middle ground for polarized discourses—a middle ground not based in
209 neutrality, but in tension.

210 This middle ground, or better, this *synthetic middle*, opens sites where both sides might come to
211 mutual understanding. In this case, the concerns with subjectivity and reality suggests that the
212 polarized positions share investments in gender as an external phenomenon. In other words, both
213 sides appear to agree that gender needs to be grounded in some way.

214 This insight, in fact, accords with what some Trans Studies scholars argue about the conceptualization
215 of gender, and how defining gender as internal causes more harm than good, creating the potential for
216 transphobic discrimination (Amin, 2022; Chu, 2024). According to Amin (2022), for example, the
217 internalization of gender identity creates a domino effect where gender is increasingly divorced from
218 bodily expression. In this situation, normative genders become both default and invisible—unmarked
219 as such. As a result, any visibly non-normative genders have a higher likelihood of experiencing
220 discrimination and stigmatization for their visible difference. Such stigmas are borne out in public
221 life where, for example, masculine gay men are more likely to participate and achieve prominence
222 over gay men who are visibly non-normative or gender-bending.

223 5 Conclusion

224 In the polarized gender discourse, the liberal investment in "reality" and the conservative investment
225 in "subjectivity" concentrate internal tensions within their respective sides, which language modeling
226 and close reading can bring to the surface. Rather than aim for objectivity, then, this methodology
227 locates sites of overlap within tension and conflict.

228 Like Konya et al. (2025), who use language modeling to facilitate dialogue between Israeli Jews
229 and Palestinians, this paper makes the strong case for language modeling as a useful tool in the
230 current landscape of polarized discourse. Unlike that work, however, which seeks a common ground

231 between opposing groups, this project approaches internal tension within groups as potential sites
232 of intersection. It makes the case for turning toward, rather than away from, social bias in language
233 forms—forms which might be operationalized for political ends.

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